Mu-Ruei Tseng

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Research Profile

Computer Science Ph.D. student working on speech and audio machine learning, streaming voice conversion, speaker anonymization, accent robustness, and efficient neural inference. I build deployable speech systems that combine representation learning, low-latency generation, privacy preservation, and rigorous evaluation.

Education

Texas A&M University, Ph.D. in Computer Science; Advisor: Prof. Ricardo Gutierrez-Osuna — Oct 2025 – Present
Texas A&M University, M.S. in Computer Science, GPA: 4.0/4.0 — 2023 – 2025
Hong Kong University of Science and Technology, B.S. in Computer Science and Mathematics; Minor in Robotics 2017 – 2021

Research Experience

Graduate Research Assistant, PSI Lab, Texas A&M University — Aug 2025 – Present
Speech/audio ML, streaming voice generation, speaker privacy, accent robustness

- Developed **VOSSA**, a streaming speaker representation framework for voice conversion that removes a separate speaker encoder and reduces model size by 19 percent; first-author paper at Interspeech 2026.
- Co-developed **TVTSyn**, an ICLR 2026 streaming voice conversion and anonymization system using content-synchronous time-varying timbre representations.
- Contributed to **DECRA**, an Interspeech 2026 real-time speech anonymization system with dynamic emotion control from continuous valence-arousal trajectories.
- Building accent-robust speech models using frozen speech foundation models, class-incremental learning, imbalance-aware optimization, and cross-corpus domain alignment.
- Built evaluation pipelines for speech generation and anonymization using WER, EER, speaker similarity, NISQA-MOS, pitch/formant analysis, ABX tests, MOS, and listener studies.

Graduate Research Assistant, PSI Lab, Texas A&M University — Feb 2024 – Jul 2025
Multimodal physiological ML, time-series modeling, clinical prediction

- Designed deep learning models for glucose prediction and hypoglycemia detection from ECG, PPG, EDA, and CGM signals.
- First-authored **ECGluFormer**, a 410K-parameter Transformer for ECG-based glucose prediction with 8.2 ms average inference time per 5-minute segment on an RTX 3090.
- Published first-author work on ECG beat-ensemble modeling for hypoglycemia prediction in the *Journal of Diabetes Science and Technology.*

Software Engineer, Neurobit Technologies — Jun 2022 – Jul 2023
Real-time eye tracking, clinical ML software, signal processing

- Built a real-time gaze estimation model running at more than 30 FPS with mean angular error below 5 degrees.
- Integrated ML inference and signal-processing pipelines into C Sharp/.NET clinical software for gaze, fixation, OKN, and vHIT examinations.
- Deployed real-time eye-movement analysis software for nystagmus assessment and automated clinical reporting.

Selected Publications

VOSSA: Voiceprint Optimization for Streaming Speech Architectures. Interspeech 2026. **Mu-Ruei Tseng**, Waris Quamer, Ghady Nasrallah, Ricardo Gutierrez-Osuna
TVTSyn: Content-Synchronous Time-Varying Timbre for Streaming Voice Conversion and Anonymization. ICLR 2026. Waris Quamer, **Mu-Ruei Tseng**, Ghady Nasrallah, Ricardo Gutierrez-Osuna
DECRA: Dynamic Emotion Control for Real-time Speech Anonymization. Interspeech 2026. Ghady Nasrallah, Waris Quamer, **Mu-Ruei Tseng**, Ricardo Gutierrez-Osuna
ECGluFormer: Glucose Prediction From ECG via Multi-Loss, Transformer-Based Aggregation. IEEE BHI 2025. **Mu-Ruei Tseng**, Ricardo Gutierrez-Osuna

Skills

Research: speech/audio ML, voice conversion, speaker anonymization, accent classification, streaming neural architectures, speaker representation learning, speech foundation models, continual learning, multimodal time-series modeling.
Engineering: Python, PyTorch, TensorFlow, scikit-learn, C Sharp, C++, Linux, Git, Docker, real-time inference, model evaluation.